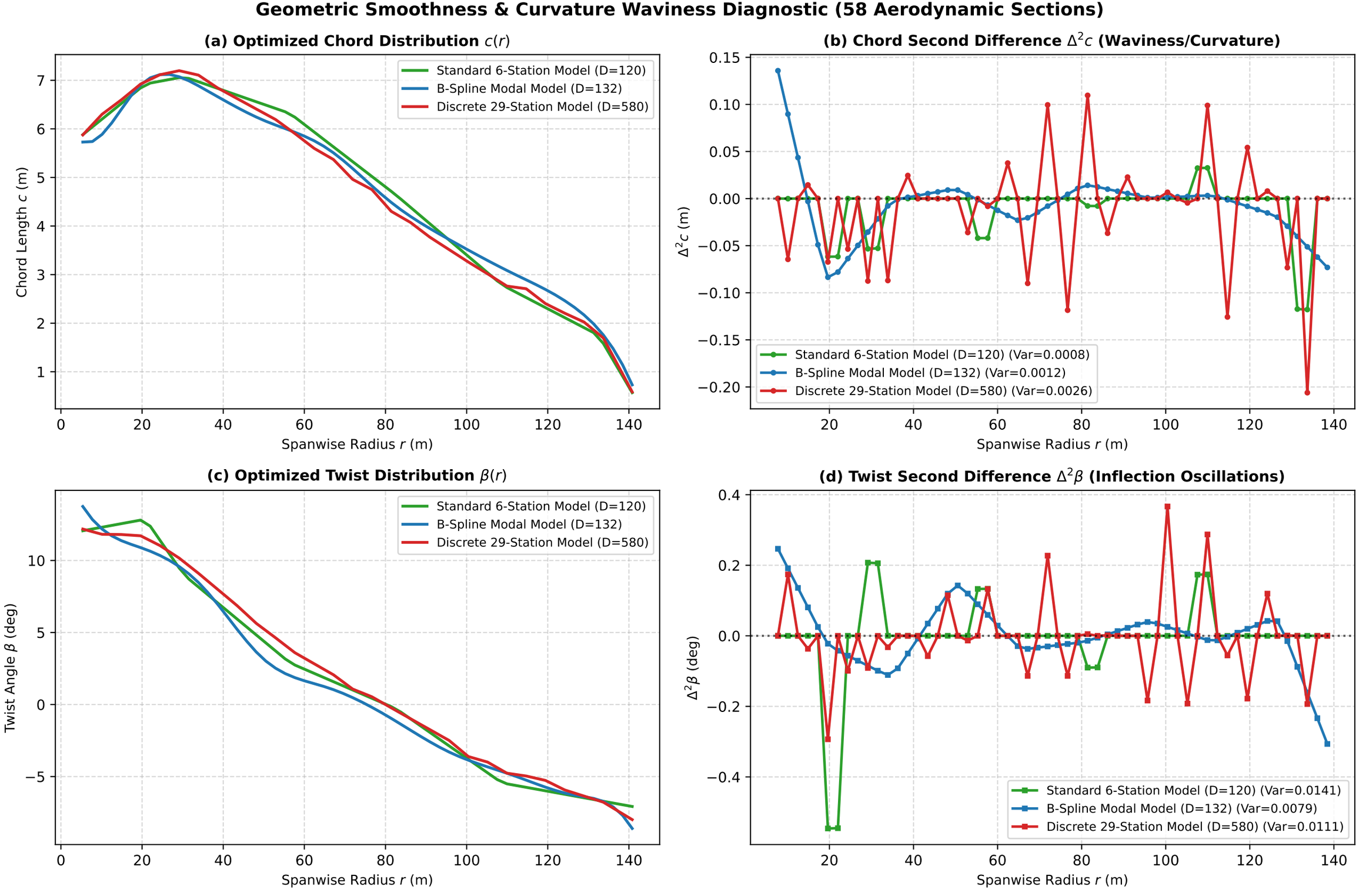
Geometric Smoothness & Curvature Waviness Diagnostic (58 Aerodynamic Sections)
(a) Optimized Chord Distribution $c(r)$
Chord Length c (m)
Spanwise Radius r (m)
Standard 6-Station Model (D=120)
B-Spline Modal Model (D=132)
Discrete 29-Station Model (D=580)
(b) Chord Second Difference $\Delta^2 c$ (Waviness/Curvature)
$\Delta^2 c$ (m)
Spanwise Radius r (m)
Standard 6-Station Model (D=120) (Var=0.0008)
B-Spline Modal Model (D=132) (Var=0.0012)
Discrete 29-Station Model (D=580) (Var=0.0026)
(c) Optimized Twist Distribution $\beta(r)$
Twist Angle β (deg)
Spanwise Radius r (m)
Standard 6-Station Model (D=120)
B-Spline Modal Model (D=132)
Discrete 29-Station Model (D=580)
(d) Twist Second Difference $\Delta^2\beta$ (Inflection Oscillations)
$\Delta^2\beta$ (deg)
Spanwise Radius r (m)
Standard 6-Station Model (D=120) (Var=0.0141)
B-Spline Modal Model (D=132) (Var=0.0079)
Discrete 29-Station Model (D=580) (Var=0.0111)